

UNC STUDENT RESEARCH DAY

School of Medicine

Presented by the

John B. Graham Medical Student Research Society

DATE	LOCATION	HOST
Nov 2, 2026	Chapel Hill, NC	UNC School of Medicine

RESEARCH • DISCOVERY • MENTORSHIP

About the Research Society

Established in 1987, the John B. Graham Medical Student Research Society recognizes and promotes the research efforts of the medical student body at the University of North Carolina School of Medicine in basic science, public health, and clinical sciences.

Students who participate in the annual Student Research Day, an event dedicated to showcase the research projects of medical students, are inducted every year to the Society. Throughout the year, members exchange ideas and share their experiences about conducting research. In addition, the Society serves to encourage collaboration with faculty to promote productive research opportunities for students.

School of Medicine

JOHN B. GRAHAM
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MEDICAL STUDENT RESEARCH SOCIETY

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Research Day Schedule

Monday, November 2nd, 2026

Location: Roper Hall (150 Medical Drive, Chapel Hill, NC 27514)

1:30 – 2:40 Keynote Speaker — Robert M. Califf, MD, MACC — Active Learning Theater

2:50 – 3:50 Networking Hour & Appetizers — Roper Hall Lobby

3:55 – 4:55 Poster Presentations

Medical Education — Lower level

Basic Science — 1st floor

Public Health — 2nd floor

Clinical Science — 3rd, 4th, 5th, and 6th floor

5:10 – 6:10 Oral Presentations

Medical Education — Room 4302

Clinical Science — Room 4310

Public Health — Room 5302

Basic Science — Room 5310

6:25 – 7:00 Dinner, Society Induction, Awards — Active Learning Theater (Reserved for student presenters, judges, and faculty guests)

Roper Hall Map

Ground Floor

Medical Education Posters — # 001–017

1st Floor

Active Learning Theater; Basic Science Posters — # 101–129

2nd Floor

Public Health Posters — # 201–229

3rd Floor

Clinical Science Posters — # 301–326

4th Floor

Clinical Science Posters — # 401–426

Room 4302: Medical Education Oral Presentations

Room 4310: Clinical Science Oral Presentations

5th Floor

Clinical Science Posters — # 501–526

Room 5302: Public Health Oral Presentations

Room 5310: Basic Science Oral Presentations

6th Floor

Clinical Science Posters — # 601–621

Keynote Speaker

Robert M. Califf, MD, MACC

Former Commissioner, U.S. Food and Drug Administration

We are honored to welcome Dr. Robert M. Califf as the keynote speaker for the 2026 Student Research Day. Dr. Califf is an internationally recognized cardiologist and leader in clinical research who served twice as Commissioner of the U.S. Food and Drug Administration. He was the founding director of the Duke Clinical Research Institute and has led landmark efforts in clinical trials, health outcomes research, and the translation of scientific evidence into clinical practice and public health policy.



Student Research Day Team

2026-2027

Co-Presidents of John B. Graham Research Society

Nina Westcott • Benjamin Lau • Maxwell Finkelstein

Vice-Presidents of Student Research Day

Jimmy Tabet • Simon Ellington • Rayaan Bashir • Allie Rosenstein • Sneha Makhijani

VP of Advertising and Outreach

CJ Tyson

For any questions about Student Research Day logistics, please contact any member of the Student Research Day team or JBGResearchSociety@med.unc.edu.

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Max Finkelstein — maxwell_finkelstein@med.unc.edu

Benjamin Lau — ben_lau@med.unc.edu

Oral Presentations

5:10 – 6:10 pm

Public Health — Room 5302

Priya Lakshmi Vadlamudi: Does Poverty Income Ratio Modify the Relationship Between Blood Metal Exposure and Gallbladder Pathology, NHANES-2021-2023

Courtney Canter: Packing Heat: school shooting preparedness, safety investment, and everyday threats in south Florida schools

Kareem El Saleh: Relationship between educational system infrastructure, presence of cardiologists and cardiovascular mortality in North Carolina

Clinical Science — Room 4310

Nathanael Haynes: The Impact of Calculated Panel Reactive Antibody on Immunologic, Patient, and Graft-level Kidney Transplant Outcomes: A Paired Institutional and National Analysis

Annaliese OuYang: Effects of Cardiac Rehabilitation Programs on Atrial Fibrillation

James Hardy: Improving Cystic Fibrosis-Related Chronic Rhinosinusitis and Nasal Nitric Oxide in the Era of CFTR Modulators

William Newman: Mean and diastolic pulmonary artery pressures predict right ventricular recovery after percutaneous pulmonary artery thrombectomy for acute pulmonary embolism

Basic Science — Room 5310

Dakota Batchek: Characterizing Critical Amino Acid Sites in DENV4 E Protein for Universal Vaccine Design

Henry Forbes Fowler: Sorafenib plus Cystine and Methionine Deprivation Diet Induces Ferroptosis in vitro and in vivo Preclinical Meningioma Models

Jordan Tran: Exacerbation of radiation-induced acute intestinal injury in Rag2-deficient mice

Justine Hill: Physiological arousal as a predominant source of individual differences in functional brain networks

Medical Education — Room 4302

Jaspreet Bhutani: Self-Compassion and Attuned Communication: A Pilot Curriculum for First-Year Medical Students

Layan Shahrour: From risk prediction to risk understanding: generative AI as a clinician-supervised communication layer for cardiac surgery

Sanya Shah: Creation of an Accessible Simulation-Based Curriculum for Pediatric Airway Foreign Body Management in Tanzania

Oral Presentation — Public Health

1. Does Poverty Income Ratio Modify the Relationship Between Blood Metal Exposure and Gallbladder Pathology, NHANES-2021-2023

Priya Vadlamudi[1,2], Rohan Chatterjee[2], Yifei Xu[2], Matthew Lamb[2]

1: University of North Carolina School Of Medicine

2: Mailman School of Public Health, Columbia University

Background: Heavy metal exposure is linked to gallbladder pathologies, yet the role of sociodemographic factors remains under-researched. This study evaluated whether heavy metals are associated with gallbladder pathology, whether age, sex, and smoking confound these associations, and if the Poverty-Income Ratio (PIR) modifies them.

Methods: We analyzed 5,014 adult participants from the 2021–2023 NHANES cohort. Exposures included blood concentrations of Cadmium, Lead, Mercury, Selenium, and Manganese. Outcomes were self-reported gallstones and history of cholecystectomy. Using SAS, survey-weighted logistic regression models were fit, adjusting for age, sex, and smoking, with PIR evaluated as an effect modifier.

Results: Females reported higher rates of gallstones (14.64%) and surgery (16.08%) than males (6.34% and 6.47%). In crude models, Cadmium significantly increased the risk for gallstones (OR: 1.16) and surgery (OR: 1.20). After adjusting for age, sex, and smoking, most metal associations attenuated, though Lead demonstrated a significant inverse relationship with gallbladder surgery (aOR: 0.54, 95% CI: 0.36–0.82, p = 0.0054). Interaction models assessing PIR as an effect modifier were non-significant (p > 0.05).

Conclusion: Heavy metal associations with gallbladder outcomes are heavily confounded by age, sex, and smoking. The inverse association for Lead likely reflects systemic healthcare barriers rather than a biological protective effect. The lack of PIR interaction indicates consistent metal-exposure associations across income levels, emphasizing the need to evaluate environmental exposures through a broader socioeconomic lens in preventing gallbladder pathology such as cholelithiasis or requirement of cholecystectomy.

Presenter Bio

Priya Vadlamudi

Priya Vadlamudi is an MS4 at the UNC School of Medicine and earned an MPH from the Mailman School of Public Health. She is applying into general surgery and is interested in public health, surgical outcomes research, and quality improvement.



Oral Presentation — Public Health

2. Packing Heat: school shooting preparedness, safety investment, and everyday threats in south Florida schools

Courtney Canter

In March 2018, just three weeks after a gunman killed 17 people at Marjory Stoneman Douglas High School (MSDHS) in Parkland, Florida, the state legislature passed SB 7026, a sweeping overall of school safety policy in the state of Florida. Schools were now obligated to follow strict safety measures, including: redesigning entrances to produce a single-point entry system, mandated security personnel-to-student ratios, fenced campus perimeter and monthly code red drills (previously yearly), among many others. These changes further entrenched well established inequities across Florida schools. This dissertation examines how “school safety” is understood, practiced, and contested in south Florida schools.Participant observation, the core qualitative methodological approach in anthropology, was the primary framework informing data collection and analysis. Primary data collection included three years (2021-2024) of ethnographic field work and participant observation in K-12 public and private schools across Miami-Dade and Broward County. Primary data collection also included 75 in-depth, semi-structured interviews and 5 focus groups with administrators, teachers, students, parents, security guards, and support staff working in Miami-Dade and Broward County. Secondary data collection included archival research utilizing school district databases, county and state demographic databases, online newspapers and televised news broadcasts. This project shows that school community members diverge sharply in their assessments of risk and the utility of safety measures. Black educators and families, whose everyday experiences with gun violence differ markedly from those of white and Latino counterparts, often view post-Parkland reforms as misaligned with the more pervasive forms of harm shaping their lives. At the same time, parents in wealthier communities marshal private resources to enhance school security, further widening disparities. Across contexts, the affective experience of preparedness—anxiety, resignation, skepticism, humor—reveals the emotional labor required to inhabit an anticipatory stance toward violence that may never materialize.Rather than treating “safety” as self-evident, this dissertation shows how safety is a historically contingent and socially differentiated value shaped by racialized experiences of gun violence, unequal resource landscapes, and the neoliberal logics of preparedness.

Presenter Bio

Courtney Canter

Courtney Canter completed her PhD in Anthropology in Fall 2025 and is a current third-year medical student. Her dissertation focused on school shooting preparedness in Florida K-12 schools following the 2018 mass shooting at Marjory Stoneman Douglas High School in Parkland, Florida. In addition to gun violence, she also has strong research interests in health care payment and delivery reform, Medicaid, value-based payment arrangements, and health system financing. She grew up in small-town North Carolina, has two toddler sons and two Rottweilers, and plans to pursue a combined Internal Medicine-Pediatrics residency.



Oral Presentation — Public Health

4. Relationship between educational system infrastructure, presence of cardiologists and cardiovascular mortality in North Carolina

Kareem El Saleh, Jacob Pierce, Rick Stouffer

Background: Cardiovascular disease (CVD) is the leading cause of death in North Carolina, and access to cardiology care is unevenly distributed. Determinants of cardiologist supply beyond urbanicity and socioeconomic factors are poorly understood. We examined whether local educational system infrastructure is associated with cardiologist distribution and CVD mortality across the state's 100 counties.

Methods: In this retrospective ecological study, cardiologist density, age-adjusted CVD mortality (2022–2024, CDC Interactive Atlas), and county covariates were assembled from federal and state sources. A novel educational composite (EC) that combined nine school- and college-related indicators was developed. Associations were evaluated using multivariable linear and logistic regression (mortality log-transformed), with FDR correction and spatial models where residual autocorrelation was present.

Results: The median density of cardiologists in NC was 1.3/100,000, 44 counties had no cardiologist and cardiologist density was inversely associated with CVD mortality. EC scores varied widely across NC counties (0.0529, IQR [-0.334, 0.306] and higher EC scores were associated with greater cardiologist density and lower CVD mortality. When counties were divided into quartiles based on EC, cardiologist density was 8.1, 2.9, 1.0 and 0.6 and CVD mortality (per 100000) was 153, 182, 193 and 210. The percentage of 11th-grade students who achieved a composite score of 17 or higher on the ACT was the most robust educational system predictor of CVD mortality.

Conclusions: In NC counties, stronger local educational infrastructure was independently associated with greater cardiologist supply and lower CVD mortality, persisting after adjustment for socioeconomic status, rural vs urban and spatial dependence.

Presenter Bio

Kareem El Saleh

Kareem is a second-year medical student at the UNC School of Medicine. His research with Dr. Stouffer examines how educational and socioeconomic infrastructure shapes the geographic distribution of cardiologists and cardiovascular mortality across North Carolina's 100 counties. He also serves as Student Director of the LMSA/PHS Cardiology Clinic, a free monthly clinic providing cardiovascular care to a largely Spanish-speaking, underserved community in Carrboro. This work has deepened his interest in cardiology and in reducing disparities in cardiovascular care.



Oral Presentation — Clinical Science

1. The Impact of Calculated Panel Reactive Antibody on Immunologic, Patient, and Graft-level Kidney Transplant Outcomes: A Paired Institutional and National Analysis

Nathanael Haynes, MS; Brian Shaw, MD; Jackie Henson, MD; Annette Jackson, PhD; Stuart Knechtle, MD

Background: High calculated panel reactive antibody (cPRA) kidney transplant (Ktx) patients historically faced barriers to transplantation. The Kidney Allocation System (KAS) elicited a four-fold increase in Ktx rate for high-cPRA (≥99%) patients. Little is known regarding escalating cPRA and risk of acute rejection (AR), as well high-quality graft allocation. This paired institutional and national analysis investigated cPRA as a predictor of AR post-KAS and discussed allocation trends across sensitization.

Methods: This retrospective cohort utilized institutional and United Network for Organ Sharing (UNOS) data post-KAS. All adult (age≥18) Ktx patients at Duke University Hospital and nationally were included (n=1,429, n=211,719 respectively). Primary outcomes included one-year biopsy-proven AR, graft survival, and patient survival. Secondary outcomes included the relationship between Kidney Donor Profile Index (KDPI) and cPRA.

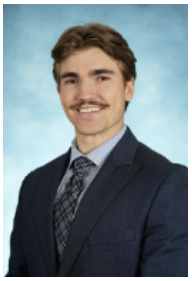
Results: Institutionally, biopsy-proven AR (p=0.46), patient survival (p=0.70), and graft survival (p=0.80) did not differ by cPRA. Kidney quality, assessed via KDPI, varied by cPRA (p<0.001, R2=0.03) with lowest mean KDPI in high-cPRA patients (p<0.001). Nationally, incidence of AR differed by cPRA (p<0.001). In unadjusted models, >98% cPRA demonstrated an odds ratio of 1.57 (p<0.001) for AR. Patient survival (p=0.63) and graft survival (p=0.37) did not vary by cPRA.

Conclusion: Post-KAS, highly sensitized Ktx patients demonstrated favorable patient and graft survival with comparable immunologic risk. These findings may be driven by the preferential allocation of high-quality donor grafts, or through advancement in desensitization protocols that are heterogeneously implemented at major centers. Future prospective studies should investigate the center-specific role of desensitization on high-cPRA immunologic risk.

Presenter Bio

Nathanael Haynes

Nathanael Haynes is a second-year medical student with interests in surgical outcomes research and abdominal transplantation. Specifically, Nathanael's research focuses on clinical science questions surrounding highly sensitized kidney transplant patients, high BMI transplantation, and the use of novel preservation strategies to optimize post-transplant outcomes. He would like to thank his clinical mentors Dr. Stuart Knechtle, Dr. Brian Shaw, and Dr. Koyal Jain for their support of this work.



Oral Presentation — Clinical Science

2. Effects of Cardiac Rehabilitation Programs on Atrial Fibrillation

Annaliese OuYang, BA; Songcheng Zhuo; Quefeng Li, PhD; Lindsey Rosman, PhD; Anthony J Mazzella, MD

Background: Cardiac rehabilitation (CR) improves quality of life, reduces readmissions, and reduces major adverse cardiovascular events for patients with qualifying diagnoses yet its effects on atrial fibrillation (AF) burden is unknown.

Methods: Patients with cardiac implantable electronic devices (CIED) capable of detecting daily AF burden were identified from the University of North Carolina Cardiovascular Device Surveillance Registry from 2015-2025. Cardiac rehabilitation attendees and non-attendees were balanced on demographics, device type, medical comorbidities, procedural history, baseline AF burden, and baseline physical activity (PA) using propensity-score overlap weights. Generalized estimating equations (GEE) measured AF presence per 30-day interval over one year of follow-up.

Results: Of 18,857 patients, 1,692 met eligibility criteria (103 CR, 1,589 non-CR). In the overlap-weighted GEE, CR significantly modified the AF trajectory (overall CR effect: $Chi^2 = 13.36$, df = 4, p = 0.0096; trajectory shape: $Chi^2 = 9.56$, df = 3, p = 0.023). The predicted probability of AF within a 30-day interval declined among CR participants to a nadir of 21.1% at day 105, versus 34.8% in non-CR participants [absolute risk difference –13.7% (95% CI −21.5 to −5.9; $p < 0.001$), 39% relative reduction] prior to gradual loss of effect over the ensuing eight months.

Conclusion: Participation in CR after CIED implantation is associated with significantly decreased probability of device-detected AF greatest in the early months after CR completion, attenuating to non-significance by 8 months. These findings demonstrate a possible benefit of CR on AF episodes. Further study is needed to determine clinical significance of these findings.

Presenter Bio

Annaliese OuYang

Annaliese is an MS2 from Central New York and a graduate of Dartmouth College. Her gap years as a medical assistant in outpatient cardiology sparked her appreciation for the breadth and clinical efficacy of cardiac interventions for optimizing patient outcomes. She serves as a Lead for the Scientific Sessions Task Force of the American Society of Echocardiography, Co-President of the UNC SOM Cardiology Interest Group, and Co-Director of SHAC Bridge to Care, and she is a recipient of the 2026 Alumni Loyalty Fund Scholarship.



Oral Presentation — Clinical Science

3. Improving Cystic Fibrosis-Related Chronic Rhinosinusitis and Nasal Nitric Oxide in the Era of CFTR Modulators

James M. Hardy, BS; Katrina Bootes, BS; Nithya Gurumurthy, BS; Ismail Khaderi, BS; Meghan Norris, PA; Charles S. Ebert, MD; Brent A. Senior, MD; Adam J. Kimple, MD, PhD

Background: Chronic rhinosinusitis (CRS) is found in almost all patients with cystic fibrosis. Both conditions are marked by a diminished ability to maintain normal concentrations of nasal nitric oxide (nNO), a vasodilator with protective immune function. Cystic fibrosis transmembrane conductance regulator (CFTR) modulator therapy is efficacious in improving CRS symptoms in people with cystic fibrosis (PwCF). Recent work indicates that this therapy also increases measured nNO levels.

Methods: In a prospective cross-sectional study, nNO levels were measured in 24 participants including PwCF and healthy controls. SNOT-22 quality-of-life scores and Lund-Kennedy nasal endoscopy scores were collected for all participants with CF. ANOVA tests compared groups for significance and Spearman rank order correlated nNO with modulator duration and symptom scores.

Results: nNO concentrations in PwCF on modulator therapy (n = 10) were significantly higher than in those who did not receive modulators (n = 4) (P = 0.0163) and were comparable to healthy controls (n = 10) (P > 0.999). Duration of modulator treatment correlated with a higher concentration of nNO (P = 0.0038). In PwCF, higher nNO was associated with a lower total endoscopy score (P = 0.0099). Modulator use showed no significant correlation with any SNOT-22 symptom measures.

Conclusion: CFTR modulators increased nNO concentrations in PwCF. nNO concentration also increased over the course of years of treatment and correlated with observed CRS disease severity.

Presenter Bio

James M. Hardy

James grew up in Wilmington, North Carolina, and attended Davidson College, graduating in 2023 with a degree in chemistry. Prior to medical school, he spent a year in Asheville on the Substance Use Disorder Taskforce at MAHEC and a year in Washington, DC, as an EMT in the 911 response system. His current clinical interests are otolaryngology and addiction medicine. Outside of academics, he enjoys playing volleyball and piano, reading his ever-expanding Stephen King collection, and seeing as much live music as possible.



Oral Presentation — Clinical Science

4. Mean and diastolic pulmonary artery pressures predict right ventricular recovery after percutaneous pulmonary artery thrombectomy for acute pulmonary embolism

William Newman, Jeffrey Diamond, Jacob Pierce, George Stouffer

Background: In patients with acute pulmonary embolism (PE), percutaneous pulmonary artery thrombectomy (PPAT) rapidly reduces thrombus burden resulting in improvements in pulmonary artery (PA) pressures. Currently, there is no consensus on the amount of thrombus that needs to be extracted for a successful procedure. The aim of this study was to evaluate the association between peri-procedural hemodynamic parameters and normalization of RV size and function.

Methods: Retrospective, observational study of 147 consecutive patients with intermediate- or high-risk acute PE who underwent PPAT at the University of North Carolina at Chapel Hill between 2016 and 2026. Pre- and post-procedural invasive hemodynamic parameters were compared using Student's t-test and analysis of variance, as appropriate.

Results: The mean age was 62 ± 18 years, 45% were females, mean BMI was 34.3 ± 8.7 and 39% were black. Pre-procedure heart rate was 100 [87, 116] bpm, PA mean pressure was 31 [26, 39] mm Hg, RV/left ventricular (LV) ratio was 1.09 ± 0.27 and 69% of patients had an RV/LV ratio > 0.9. On follow-up echocardiography (n = 126), 84 (67%) patients had normal RV size and function. PA mean pressure (p=0.038) and PA diastolic pressure (p=0.0037) measured at the end of the procedure predicted RV recovery. In contrast, neither heart rate, post-procedure PA systolic pressure nor pre-procedural hemodynamics were associated with RV recovery.

Conclusion: Post-procedural PA mean and diastolic pressure measurements predicted RV recovery and thus could potentially be used to determine when sufficient thrombus has been extracted to successfully conclude PPAT.

Presenter Bio

William Newman

Will is an MS2 from Raleigh, NC. He graduated from Davidson College in 2023 with a major in biology. Will is interested in internal medicine and cardiology. During his free time, he enjoys fishing and anything outdoors!



Oral Presentation — Basic Science

1. Characterizing Critical Amino Acid Sites in DENV4 E Protein for Universal Vaccine Design

Dakota Batchek, Z Beau Reneer, PhD, Ralph Baric, PhD

Background: Dengue viruses have 4 distinct serotypes with dengue virus 4 (DENV4) having the least phylogenetic diversity. DENV4 has 6 different genotypes with sequence variation in the E protein domain which impacts antibody recognition. Identifying which naturally variable amino acid sites are critical for antibody binding is pivotal for designing an antigen for vaccines that cross-reacts with all DENV4 strains.

Methods: Using Computationally Optimized Broad Reactive Antigen (COBRA) methodology, we created hybrid E protein antigens that incorporated significant amino acids from the DENV4 IIb wildtype (WT) genotype, genotype I, and genotype III. We generated seven single- or multi-site mutants at positions 200, 202, and 233 of the DENV4 E protein domain II. Neutralization assays were performed using monoclonal antibodies (mAbs) (D4-126, D4-131, EDE1-C10) to determine how mutations at these positions affect antibody binding.

Results: Position 202 emerged as critical for genotype-specific antibody recognition: the K202E mutation eliminated D4-126 binding [reciprocal half maximal effective concentration (EC50) >10ng/uL) while maintaining D4-131 binding (EC50 = 0.19ng/uL). Positions 200 and 233 showed moderate effects with some interactions: the K202E-Y233H double mutant partially recovered D4-126 binding (EC50 = 0.084ng/uL) compared to K202E alone. Critically, all variants maintained full cross-reactive mAb EDE1-C10 binding (EC50 = 0.062-0.19ng/uL).

Conclusion: Positions 200, 202, and 233 are variable sites that drive genotype-specific antibody escape. However, cross-reactive antibodies remain functional against all variants. These findings suggest universal DENV4 vaccines should prioritize these three amino acid sites to account for more genotype breadth while also inducing cross-reactive antibody responses.

Presenter Bio

Dakota Batchek

Dakota is an MS2 at the UNC School of Medicine from High Point, North Carolina. He graduated from UNC in 2025 with a degree in Nutrition Science and Research from the Gillings School of Global Public Health. He is interested in surgery and enjoys doing research across many fields. In his free time, he loves watching the Philadelphia Eagles. Go Birds!



Oral Presentation — Basic Science

2. Sorafenib plus Cystine and Methionine Deprivation Diet Induces Ferroptosis in vitro and in vivo Preclinical Meningioma Models

Henry Forbes Fowler, Olivia Gilbert, Asish Madarapu, Dominique Higgins

Meningiomas are the most prevalent primary central nervous system (CNS) tumor, arising from arachnoid cap cells. Though the vast majority of these tumors are benign, recurrence rates vary widely. While rare, atypical or malignant meningiomas can occur and are associated with lower recurrence-free survival rates, often with debilitating neurological deficits. There are no universally effective treatment options for aggressive and recurrent meningiomas, as these often exhibit radioresistance. Ferroptosis has emerged as a promising therapeutic approach for a variety of cancers, including malignant gliomas. This study explores the effect of ferroptosis induction in meningioma cells in vivo and in vitro, aiming to determine if ferroptosis is a viable therapeutic avenue for meningiomas. Four separate meningioma cell lines were treated with varying concentrations of sorafenib, a tyrosine kinase inhibitor that has also been shown to induce ferroptosis, in control (DMEM) or cystine-methionine-deprived (CMD) media, and viability was assessed. All cell lines demonstrated sensitivity to sorafenib, with sorafenib exhibiting synergy with CMD in a dose- and time-dependent manner in vitro. In vivo dose-response to sorafenib was assessed in 20 mice before performing subsequent sorafenib and CMD trials in 20 additional mice. Intracranial tumors were monitored using bioluminescent imaging twice weekly. Sorafenib plus CMD diet reduced overall tumor growth rates but did not exhibit clear synergistic effects in vivo. Together, this preclinical work suggests sorafenib and other ferroptosis inducing agents may be viable avenues for intracranial meningioma treatment.

Presenter Bio

Henry Forbes Fowler

Forbes Fowler is a MS2 at the UNC School of Medicine from Asheville, NC. He graduated in 2024 with a BS in Biology from UNCA. Building on basic science work done as an undergraduate, He is pursuing research interests in neurobiology and neurosurgery.



Oral Presentation — Basic Science

3. Exacerbation of radiation-induced acute intestinal injury in Rag2-deficient mice

Jordan Tran, Lucy Driver, Lixia Luo, Chang-Lung Lee

Background: Gastrointestinal acute radiation syndrome (GI-ARS) is a life-threatening illness that occurs following high-dose abdominal irradiation. However, no FDA-approved countermeasures exist. A major barrier to therapeutic development is our limited understanding of how radiation-induced immune responses regulate intestinal tissue regeneration after injury. This project aimed to investigate the modulatory role that mature lymphocytes play in post-irradiation epithelial cell death and regeneration.

Methods: Rag2-/- mice, which lack mature T and B cells, and littermate Rag2+/- controls received 13.5 Gy sub-total-body irradiation (SBI). Intestinal regeneration was assessed by BrdU incorporation in crypt cells and expression of stem cell marker Olfm4 at 96 hours post-SBI, and by Clusterin, a marker of regenerating epithelium, at 72 hours. Crypt apoptosis was quantified by cleaved caspase-3 staining at 4 hours. Mature T versus B cells were depleted using Cre-mediated diphtheria toxin-A.

Results: Rag2-/- mice had significantly reduced survival compared to controls. Regeneration was impaired at 96 hours post-SBI, with significantly fewer BrdU-labelled crypt cells and reduced Olfm4 expression, and Clusterin induction was blunted at 72 hours. Rag2-/- mice also showed a two-fold increase in cleaved caspase-3 positive crypt cells at 4 hours. Depletion of individual subsets identified mature T cells as the primary lymphocyte population suppressing radiation-induced crypt apoptosis.

Conclusion: Together, these results demonstrate that Rag2 deficiency, particularly the lack of mature T cells, exacerbates early apoptosis of crypt cells and impairs intestinal regeneration following severe radiation injury in mice, providing novel insight into the intersection of radiation-induced immune responses and epithelial cell damage and recovery.

Presenter Bio

Jordan Tran

Jordan Tran is a first-year medical student at the University of North Carolina School of Medicine. His academic interests include radiation oncology and immunology, with a particular focus on translating scientific discoveries into innovative therapies for patients with cancer. He hopes to pursue a career that integrates clinical care, research, and medical education.



Oral Presentation — Basic Science

4. Physiological arousal as a predominant source of individual differences in functional brain networks

Justine A. Hill, Tianye Zhai, Cole Korponay, Julia C. Welsh, Betty Jo Salmeron, Thomas J. Ross, Yihong Yang, Blaise deB. Frederick, Amy C. Janes

Characterizing individual variation in brain network organization is essential for developing fMRI-based biomarkers in precision psychiatry. We show that arousal, assessed by systemic low-frequency oscillation (sLFO) amplitude in the fMRI signal, is strongly associated with interindividual variability in functional brain network properties (e.g., R = 0.70 for default mode network (DMN)–dorsal attention network (DAN) connectivity; R = −0.63 for DMN dynamics), with replication across independent sessions and samples. These relationships persist after regressing the sLFO out of the BOLD signal and are replicated using established physiological arousal indices, indicating that arousal-related physiology significantly contributes to variability in brain network measures. Pharmacological manipulation shows that increased arousal under catecholaminergic agonism is associated with enhanced network differentiation, whereas antagonism produces the opposite effect. These findings highlight arousal-related physiological processes as a major contributor to individual differences in functional brain network properties as estimated with fMRI and indicate that catecholaminergic modulation is associated with these effects via arousal-related processes.

Presenter Bio

Justine A. Hill

Justine Hill graduated with a degree in Neurobiology from the University of Wisconsin-Madison in 2022. She went on to study co-occurring substance use and mental health disorders at the National Institute on Drug Abuse before joining the UNC MD-PhD program this fall. As a physician-scientist, Justine hopes to use computational neuroscience to improve understanding of complex, co-occurring psychiatric disorders and apply this insight to create individualized neuromodulatory treatment approaches.



Oral Presentation — Medical Education

1. Self-Compassion and Attuned Communication: A Pilot Curriculum for First-Year Medical Students

Jaspreet Bhutani, MS; Jessica Barnhill, MD, MPH; Linda Gilkerson, PhD, LSW; Shirley Thoms, MPA; Jin Park, PhD; Christine Lathren, MD, MSPH

Background: Medical students face high rates of burnout, stress, and compassion fatigue, linked to reduced capacity for patient care. Training in self-compassion and attuned communication may help students manage personal distress while maintaining their ability to care for others, yet medical schools rarely address these skills. This study examined the feasibility, acceptability and associated changes in well-being and communication outcomes for first year medical students participating in a new training, Self-Compassion and Attuned Communication.

Methods: The first of two cohorts of first-year medical students (N=31) voluntarily enrolled in 10 lunchtime sessions (6 self-compassion, 4 attuned communication) across the Spring 2026 semester; a second cohort is planned for 2027. Feasibility and acceptability were captured by weekly attendance, post-training ratings and survey responses, and group feedback sessions. Surveys captured anxiety, burnout, depression, emotion dysregulation, self-compassion, and communication skills at pre/mid/post training; changes were examined via paired-samples t-tests. Transcriptions from audio-recorded feedback sessions and open-ended acceptability responses were analyzed using thematic analysis.

Results: Attendance and acceptability were high (9.1/10 sessions average; no dropouts), with 95% reporting a positive well-being impact. Significant improvements were observed in anxiety, emotion dysregulation, self-compassion, and communication skills. Qualitative analysis identified six themes, showing students perceived a curricular gap, gained applicable skills, and wanted the program to continue.

Conclusion: This pilot demonstrates that SCAC is feasible, well-received, and associated with improvement in well-being, self-compassion, and communication skills. SC/AC skills should continue to be examined as potentially unique and valuable components of medical education.

Presenter Bio

Jaspreet Bhutani

Jaspreet Bhutani is a second-year medical student at the UNC School of Medicine. She took part in the Spring 2026 pilot of the Self-Compassion and Attuned Communication (SCAC) course and went on to lead its qualitative data analysis, a grounded theory study of student feedback. She is now leading a manuscript on the findings with Dr. Lathren. Outside research, Jaspreet co-directs Bridge to Care at the student-run free clinic Student Health Action Coalition (SHAC) and hopes to pursue a career at the intersection of medicine and health policy.



Oral Presentation — Medical Education

2. From risk prediction to risk understanding: generative AI as a clinician-supervised communication layer for cardiac surgery

Layan Shahrour, Nesreen Shahrour, Sammy Chouffani El Fassi, Issam Alzouby, Zachariah Ngan, Khar Dieng-Datta, Yeahwa Hong, Carmelo A. Milano, Thomas G. Caranasos, Manesh R. Patel

Background: Validated Society of Thoracic Surgeons (STS) models estimate operative mortality and major complications in adult cardiac surgery, but numerical estimates may not adequately communicate recovery and functional consequences. Differences in health literacy and communication may limit informed consent and shared decision making.

Methods: CardiOp-AI is a clinician-supervised generative artificial intelligence (AI) communication layer designed to complement, not replace, risk prediction. The prototype extracts information, organizes risk factors, identifies missing or uncertain data, and generates editable explanations linked to STS estimates. Patient-facing explanations contextualize probabilities by describing contributing factors, recovery implications, and mitigation strategies. The prototype was evaluated using one synthetic high-risk coronary artery bypass grafting case with end-stage kidney disease requiring dialysis, diabetes, and pulmonary hypertension.

Results: CardiOp-AI generated STS-compatible inputs and produced estimates comparable to manual STS calculation. Operative mortality was 4.5% versus 4.31% by manual assessment, and major morbidity/mortality was 15.0% versus 14.19%. Preparation time decreased from 17-20 minutes to less than 1 minute. The single synthetic case only demonstrates technical feasibility and does not establish clinical accuracy, safety, generalizability, or patient benefit.

Conclusions: CardiOp-AI illustrates a potential framework for translating validated surgical risk estimates into individualized explanations while preserving clinician responsibility for communication. Safe implementation requires clinician review, transparent linkage to validated STS estimates, auditable outputs, and monitoring for hallucinations and bias. Prospective studies should assess patient understanding as the primary outcome, with secondary assessment of recall, decisional conflict, preparedness, shared decision-making quality, clinician usability, factual corrections, and equity.

Presenter Bio

Layan Shahrour

Layan Shahrour is a medical student at the UNC School of Medicine with interests in research, health equity, and the intersection of medicine and the humanities. Her research spans clinical medicine, psychosocial health, medical AI, and health disparities, with publications in peer-reviewed journals. She is passionate about advancing patient-centered care and addressing health inequities through research, community engagement, and advocacy.



Oral Presentation — Medical Education

3. Creation of an Accessible Simulation-Based Curriculum for Pediatric Airway Foreign Body Management in Tanzania

Sanya Shah, Cody Messick, Kassandra Jade, Evani Patel, Amanda Godoi, MD, Jeffrey Black, Grace Travers, Sean Tzoucalis, Karnavaal Al-Rubayie, Joshua Wiedermann, MD, Cecilia Protas, MD

Background: Foreign body aspiration is a major cause of preventable morbidity and mortality among young children, with disproportionately high complication rates in low- and middle-income countries (LMICs). Effective management requires timely recognition and technical proficiency, yet training opportunities may be limited by shortages of specialized providers, equipment, and accessible simulation-based education.

Aims: To develop and evaluate a low-cost, scalable simulation-based curriculum for pediatric airway foreign body management in Tanzania, assessing educational outcomes, simulator and curriculum validity, and factors influencing implementation feasibility, acceptability, and sustainability.

Methods: A prospective mixed-methods study was conducted at Bugando Medical Centre and the Catholic University of Health and Allied Sciences in Mwanza, Tanzania. Otolaryngology residents and specialists participated in a hybrid curriculum consisting of online modules and hands-on training using a low-cost, 3D-printed airway simulator. Pre- and post-intervention assessments evaluated clinical knowledge, decision-making, and technical skills using a multiple-choice examination, Objective Structured Clinical Examination, and Objective Structured Assessment of Technical Skills, respectively. Surveys, validity assessments, and qualitative interviews and focus groups were also conducted.

Results: Data collection was completed during summer 2026. Analysis is pending approval from the Tanzanian National Institute for Medical Research for international data transfer, which is anticipated in October 2026. Planned analyses will evaluate changes in learner performance and characterize curriculum feasibility, acceptability, sustainability, and perceived utility.

Conclusion: This study developed and implemented an accessible simulation-based curriculum for pediatric airway foreign body management. Findings will inform refinement and potential expansion of low-cost simulation-based surgical education models in Tanzania and other LMIC settings.

Presenter Bio

Sanya Shah

Sanya is a second-year medical student at the UNC School of Medicine and a member of the Global Health Scholarly Concentration. This past summer, she spent two months at Bugando Medical Center (BMC) in Mwanza, Tanzania, with Partnerships for Global Surgery (PGS). There she implemented a low-cost simulator and training curriculum for pediatric airway foreign body management, mentored by Drs. Cecilia Protas (BMC) and Josh Wiedermann (Mayo Clinic, PGS). The project is now expanding to Ethiopia for implementation and to Mayo Clinic for further validation. Sanya's interest in global health was kindled by her early childhood in Navsari, India, and grew through her work on North Carolina's COVID-19 Vaccination Taskforce and with UNC Project Malawi. She hopes her wanderlust will take her to many more communities, where she can serve as a physician who listens first and acts thoughtfully.



Poster Presentations

Posters are categorized as Medical Education (ME), Public Health (PH), Clinical Science (CS), or Basic Science (BS).

#	Topic	Presenter	Title
001	ME	Stephen Colella	Venetoclax With Intensive Induction Chemotherapy in Newly Diagnosed Acute Myeloid Leukemia: A Focused Evidence Review
002	ME	Jessie Glanz	A Gut Feeling: Examining Fecal Microbiota Transplantation on Psychiatric Symptoms in Irritable Bowel Syndrome
003	ME	Kristen Dailey	GLP-1 Receptor Agonist Therapy and Risk of Recurrent Pancreatitis in Adults With Type 2 Diabetes or Obesity
004	ME	Lucas Boldt	Population-based Survival Analysis of Malignant-Coded Odontogenic Tumors (ICD-O-3 9310/3) in the SEER Database
005	ME	Numair Attaar	Preoperative Psychoactive and Analgesic Medication Use in Neurogenic Thoracic Outlet Syndrome
006	ME	Josh Sun	Disaster Medicine and Dermatology: A Scoping Review of Cutaneous Diseases and Dermatologists' Roles After Natural Disasters
007	ME	Eric Nungester	Title: Depression and neuroplasticity: How psilocybin is unlocking new tools of Major Depressive Disorder.
008	ME	Luke Thomas	Young Age and Meniscectomy as Independent Predictors of Contralateral ACL Retear Following Primary ACL Reconstruction
009	ME	Katherine Snyder	Gut Instincts: Translational Evidence for Fecal Microbiota Transplantation in Substance Use Disorders
010	ME	Robert Michael Rampani	Ready in Principle, Limited in Practice: North Carolina EMS Leaders' Perspectives on Advanced Practice Provider Integration
011	ME	Marlena Raney	Stones, Bones, and MTP Groans: An evaluation of Calcium Repletion Strategies in Massive Transfusion Protocols for Trauma
012	ME	Megan Boudah	Not So Sweet: Investigating the Expanded Effects of GLP-1 Agonists in Non-Diabetic Patients
013	ME	Ermina Ciric	Completeness of the Circle of Willis and 90-Day Functional Outcomes Following Acute Ischemic Stroke
014	ME	Lacey Elko	Should Stage 1 Hypertension Be Added as a Moderate-Risk Factor for Aspirin Prophylaxis in Pregnancy? An Integrative Review
015	ME	Sarah Bishop	Standing Under Pressure: Can Abdominal Compression Improve Symptoms in Adults with Postural Orthostatic Tachycardia Syndrome (POTS)?
016	ME	Philip Haire	ACLS Among the Stars: Adapting Emergency Care for Commercial Spaceflight
017	ME	Ahmed Elinbabi	Adjunctive Azithromycin in Children with Poorly Controlled Persistent Asthma: A Literature Review
101	BS	Hayley Ross	Adolescent intermittent stress causes lasting adult proinflammatory innate immune activation and loss of cholinergic neurons in association with increased alcohol drinking in adulthood

#	Topic	Presenter	Title
102	BS	Preetam Tanikella	Generating ECGs for Computational Electrophysiology Models: A Numerical Framework
103	BS	Amy Dokiburra	Senolytic therapy modulates microglial phenotype and clinical disease severity in an age-dependent model of progressive multiple sclerosis
104	BS	Nicholas Alex Rose	Algorithms for Characterizing Motion-Based Energy Harvesting Reproducibility
105	BS	Kiana Garakani	Alveolar Type II Cell-derived MMP1high basal cells promote destructive microcysts in idiopathic pulmonary fibrosis
106	BS	James Carmouche	IL-2 complexes enhance the persistence and effector functions of adoptively transferred anti-pancreatic CAR T helper 17 cells
107	BS	Juliana Tovar	The Role of Microglia in Acute Stress and Addiction in the Nucleus Accumbens Core
108	BS	Avinash Jayakumar	ASSOCIATION BETWEEN ERRORS COMMITTED DURING AUDITORY FEEDBACKTRAINING AND NONCONTACT ACL INJURY RISK
109	BS	Molly Dreher	Immune-Directed Neural Differentiation of Placental Stem Cells
110	BS	Logan Reeg	Correlating Structural Joint Degeneration with Behavioral Pain Sensitivity in Aged Murine Models of Osteoarthritis
111	BS	Aaron Claypool	Intermittent access to oxycodone self-administration and alcohol alters oxycodone intake and the reinforcing efficacy of oxycodone in a sex-specific manner
112	BS	Oviya Kalaivanan	Effect of Knockdown of a Novel DNMT3A Interactor in a Model of Huntington's Disease
113	BS	Satya Yalamanchi	Investigating the role of CRL4Cdt2 and Condensin I subunits after DNA damage
114	BS	Makayla Poindexter	Evaluation of Senescent Signatures in Osteoarthritic Chondrocytes Reveals Heterogeneity
115	BS	Kareem Haidary	Exploring the Effects of Alcohol on G9a activity and H3K9me2 levels in Stress-Related Brain Regions
116	BS	Cheyenne Sadeghi	Predicting Palbociclib Response with Single-Cell Diffusion Models
117	BS	Giuliana Lopez	Characterizing the Influence of Laminin-332 on Extracellular Matrix-Mediated Invasion in Pancreatic Ductal Adenocarcinoma Organoids
118	BS	Rahul Viswanath	Pharmacological Depletion of Retinal Mononuclear Phagocytes Is Neuroprotective in a Mouse Model of Mitochondrial Optic Neuropathy
119	BS	Eric Harris Albanese	An Ao-kay investigation:A cytological approach to the molecular role of Abnormal Oocyte (ao)
120	BS	Spencer Mohandas Maingi	Characterizing Neuronal-Glioma-T cell Crosstalk in Glioblastoma Using Spatial Transcriptomics
121	BS	Kees Heetderks	Characterizing Subcortical Speech and Language Processing Using Auditory Electrophysiology
122	BS	Bhavika Chirumamilla	Combined inhibition of PARP and BET proteins impairs DNA damage response in preclinical models of bladder cancer
123	BS	Sophia Nigrovic	Evaluation of design parameters for monosaccharide probes used in the metabolic labeling of bacterial glycans

#	Topic	Presenter	Title
124	BS	Nithya Gurumurthy	Evaluation of Environmental Exposures, Disease Severity, Quality of Life, and Olfaction in Chronic Rhinosinusitis
125	BS	Kendall Flaharty	Mapping the causal mechanisms of inherited retinal disease: agentic expansion of the Disorder Mechanisms Knowledge Base (Dismech)
126	BS	Rhys Evans	Neurofilament light chain as a prognostic marker in cerebral adrenoleukodystrophy
127	BS	Derek Lin Chien	Promoting the Genetic Stability of Human Pluripotent Stem Cell Cultures
128	BS	Dhairya Desai	Study of Antimicrobial Peptides for Peptide-DNA Conjugates as Novel Therapeutic Agents and the Effect of Cellular Stress on the Localization of Atox1
129	BS	Ashton Rierson	U6 is not an Optimal Endogenous Control for Normalization of Circulating miRNAs in Thoracic Aortic Aneurysm Disease
201	PH	Megan Liu	Delayed effects of flooding on malaria transmission in southeastern Tanzania: a district-level distributed lag analysis
202	PH	Evan Rosenberg	Support, Readiness, and Barriers to Automated External Defibrillator Drone Adoption: A Statewide Survey of Emergency Medical Services Leaders across North Carolina
203	PH	Jennifer Te Vazquez	Developing a Brief and Culturally Responsive Intervention for Disordered Eating: Adolescent and Caregiver Perspectives
204	PH	Alexandra Zaffuts	The Methodological Quality of Cost-Effectiveness Studies of Palliative Care in Non-Cancer Serious Illness: A Rapid Systematic Review
205	PH	Vishanth Palanivel	Electronic Health Record–Based Detection of Bleeding in Older Adults: Enhancing ICD-10-CM Algorithms with Chief Complaint Data
206	PH	Ryan Norris	Trends in Mental Health Counseling and Therapeutic Treatment for Depression Among Cancer Survivors
207	PH	JeMya McClendon	"A Bucket of Money Would Be Nice.”A mixed method analysis of how federal funding cuts impacted SAFEChild’s programs and operations
208	PH	Cindy Shou	Evaluating the Facilitators and Barriers to Implementation of Teacher-Delivered Transdiagnostic Mental Health Care in North Carolina Schools
209	PH	Preetha Ramachandran	A Qualitative Study of Barriers to Parental Consent in a School-Based Mental Health Intervention
210	PH	Drew Varnell	Injury Patterns of Fishing Related Injuries Presenting to the Emergency Department in the United States
211	PH	Noor Alghanem	Barriers to Breast Cancer Screening Among Arab Women: A Thematic Narrative Review
212	PH	Danielle Schmidt	Molecular Characterization of Viral Acute Gastroenteritis in Children from Southwest Guatemala
213	PH	Emma Finch	Beyond the Map: Assessing True Abortion Access in Post-Dobbs Western North Carolina
214	PH	Sophia Dhrolia	Period poverty and menstrual experiences in women living with HIV
215	PH	Elisabeth McIntosh	Shared Reading Frequency in the First Year of Life and Subsequent Screen Time and Speech Delay

#	Topic	Presenter	Title
216	PH	Chad Urquhart	Evaluation of Liver-Directed Therapies for Colorectal Liver Metastases: A National Cancer Database Study
217	PH	Sophia Goldbeck, Julie Lee	Caregiver Roles and Responsibilities in Managing the Patient Transition Home After Critical Illness: A Qualitative Analysis
219	PH	Emma Domke	Facilitators and barriers to the use of permethrin-treated lesus in Kilifi, Kenya
220	PH	Eleanor Steiner	From Root Cause to Redesign: Applying the PRECEED-PROCEED Model to Design an Integrated Pediatric Medical Home at WakeMed
221	PH	Saratu Garba	From Your Screens to Getting Vaccines: Sources of Vaccine Hesitancy Amongst Pregnant British African & Caribbean Women in London
223	PH	Sumiya Bibi	Increasing Vaccine Uptake among Black African and Caribbean Women and Children: Co-creation and Focus Groups in London, UK
224	PH	Anusha Dubey	Racial and Ethnic Differences in Drug Toxicology Testing
225	PH	Chase Jarvis	Two-Year Results of a County EMS Medications for Opiate Use Disorder (MOUD) Treatment Program
226	PH	Elizabeth Malone	Understanding how Adolescents and Young Adults use Reddit to Learn about Opill: A Qualitative Content Analysis
227	PH	Aydin Bandukwala	Understanding Suicide Risk and HIV Care Engagement Among Adolescents Living with HIV in Lilongwe, Malawi
228	PH	Connor LaRocca	Utilizing ICD Coding and Keywords in Emergency Medical Services Data to Improve COPD and Asthma Surveillance
229	PH	Darren Sanders	Factors Associated with Missed Medication Doses During Community Reentry Among NC FIT Participants
301	CS	Rachel Li	Association of lipoprotein(a) with coronary artery calcification in former professional athletes.
302	CS	Trieu-Vi Khuu	Demographic and Clinical Factors Associated with Engagement in an AI-Assisted Rehabilitation Program
303	CS	Rida Zainab	Genetic Evaluation of Pediatric Pituitary Adenomas and USP8-Related Genotype-Phenotype Correlations in Cushing's Disease
304	CS	Nandan Patel	Is Rituximab the Secret Sauce for MuSK MG Remission?
305	CS	James Petrongelli	PSMA-PET-Guided SBRT Is Associated with PSA Response in Oligometastatic Castration-Resistant Prostate Cancer: A Multi-Institutional Retrospective Analysis
306	CS	Sahana Giridharan	Underreporting of Hepatic Steatosis on Preoperative CT Scans in a Modern Hepatectomy Patient Cohort
307	CS	Kayleigh Doherty	Association of Perioperative Suzetrigine with Post-Operative Rehabilitation Therapy and Opioid Prescriptions Following Orthopedic Shoulder Surgery
308	CS	David Alexander Perez-Benitez	Demographic and Complication Differences in Distal Radius Fracture ORIF Patients Between TriNetX and PearlDiver Databases: Implications for Cross-Database Research
309	CS	Rayaan Muhammad Bashir	Graft Type and Meniscal Intervention Affect Landing Kinematics at Return to Sport After ACLR
311	CS	Sydney Morgan	Using GENO-STELLAR to predict outcomes in Klebsiella pneumoniae infection

#	Topic	Presenter	Title
312	CS	Bergen Murray	Documenting Pre-Implementation Adaptations of the Goals of Care Intervention: A FRAME-Guided Qualitative Analysis
313	CS	Joe LaVigne	Health-Related Quality of Life in Focal Segmental Glomerular Sclerosis and Minimal Change Disease: A Qualitative Study of Children and Adults to Inform Patient-Reported Outcomes
314	CS	Spandan Goel	Physiologic Ferritin Reference Ranges Increase Diagnosis and Treatment Initiation of Iron Deficiency Without Anemia
315	CS	Sophia de Araujo	Relationship Between Drug Class and Mucosal Site Involvement with Steven-Johnson Syndrome/Toxic Epidermal Necrolysis (SJS/TEN)
317	CS	Sriram Srinivasan	Where to biopsy? Time to clinical intervention and mortality associated with inpatient vs outpatient targeted liver biopsies for histopathologic diagnosis
318	CS	Phoebe Pak	A Descriptive Analysis of GLP-1 Receptor Agonist Prescribing Among Patients With Serious Mental Illness at UNC
319	CS	Nathan Mothes	Clinical Outcomes of Aspirin Continuation Versus Discontinuation Following A Bleeding Event in Older Adults Using Aspirin: A Retrospective Cohort Study
320	CS	Ian Francis Miller	Dyadic Care Transitions Following an ICU Stay for Acute Respiratory Failure: A Journey Mapping Analysis
321	CS	Jessica Wu	Hospital Wastewater Surveillance as a Way to Measure Dengue Virus Incidence in Cambodia
322	CS	Molly Haggerty	Postpartum Knee Dysfunction and Return to Activity
323	CS	Forbes Fowler	Sarcoid-like Granuloma of the Sacral Spine Mimicking a Calcified S1 Nerve Root Schwannoma
324	CS	Madelyn Perkins	The Relationship between Body Mass Index and Interoperative Blood Loss and Transfusion Rates among Patients Undergoing Liver Resection: A Retrospective Cohort Study
326	CS	Catherine Bundon	Effects of Clinical Decision Support Tool Implementation on Screening Patterns in Older and High-Risk Men
401	CS	Michael Maizel	Improving early diagnosis of Acanthamoeba keratitis using deep learning analysis of slit-lamp images
402	CS	Tifani Panek	Predictors of Initial and Repeat Shunt Revision Following Transjugular Intrahepatic Portosystemic Shunt: A Comprehensive Cohort Analysis in a Single Institution
403	CS	Sean A. Celli	Sex Differences in SNOT-22 Scores: Retrospective Review and Analysis from a Single Center Institution
405	CS	Reya Patel	Association between Mitral Annular Calcification and Elevated Coronary Artery Calcium Score
406	CS	Sahib Chandi	Confidentiality Concerns as a Barrier to PrEP Uptake Among At-Risk Populations: A Pilot Survey Study
407	CS	Claudia Barreto	Evaluation of a novel EHR-based diabetes distress screening program for adults with type 1 diabetes
409	CS	Quay Parrott	Preoperative Denosumab Exposure and Outcomes Following Spinal Arthrodesis: A Real-World Analysis of 1,232 Patients
410	CS	Stephanie Ntim	The Value of Scalp Hematomas in Predicting Risk of Traumatic Intracranial Hemorrhage

#	Topic	Presenter	Title
411	CS	Sydney Hawkins	Association Between Nutrition Security and Glycemic Control in Adults with Type 2 Diabetes Enrolled in a Food-Is-Medicine Pilot Trial at Baseline
412	CS	Emma Comadoll	Cost-Effectiveness Analysis of Adjuvant Cemiplimab Plus Surgery and Radiation Versus Surgery and Radiation Alone for High-Risk Cutaneous Squamous Cell Carcinoma
413	CS	Alexandra Helbling	Preoperative Inflammatory Markers and Anterior Cruciate Ligament Reconstruction Outcomes
414	CS	Enes Eroglu	Thirty-Day Complications Following Bariatric Surgery in ESRD Patients: A Retrospective Cohort Study in a Publicly Insured Cohort
415	CS	Rishita Chamarthi	Association Between Pre-ICU Functional Status and ICU Resource Utilization and Clinical Outcomes
416	CS	Grace Matwijec	Evaluation of Lifestyle Factors on Diabetes Distress in Emerging Adults with Type I Diabetes
417	CS	Joshua Morningstar	Increased risk of 5th Metatarsal Fractures Among Patients Receiving Denosumab Therapy: A Matched Cohort Analysis
418	CS	Tanzeel Ahmad Khan	Trends in Anti-Osteoporosis Pharmacotherapy Following Sentinel Distal Radius Fracture, 2015–2025
419	CS	Claire Pedersen	Adapting and Validating a Patient-Reported Experience Measure (PREM) tool for Caregivers of Children Hospitalized at Kamuzu Central Hospital
420	CS	Abia Ahmed	Comparison of Four Tools for Continuous Glucose Monitoring Data Analysis
421	CS	Charlie Piehl	Early Mobilization and Verticalization in Critically Ill Pediatric Populations:A Review in Extracorporeal Membrane Oxygenation, in Burns, and in Trauma
422	CS	Rishi Aghera	Identification of Cardiac Proteins Associated with the Risk of Myocardial Infarction Using UK Biobank Proteomics Data
423	CS	Alina Shcherbakova	Predicting Length of Stay for Patients Receiving Head CTs for Nontraumatic Neurological Symptoms in the Emergency Department
424	CS	Teresa Zheng	Screening and Early Detection of Vulvovaginal Chronic Graft-versus-Host Disease: A Scoping Review
426	CS	Mansi Gupta	Evaluation of DEXA Scan Use and Bone Density Outcomes in Patients with Turner Syndrome
501	CS	Rennica Huang	Improving the Study Design of an Age-Friendly Continuous Glucose Monitoring Education Clinical Trial
502	CS	Seth Eller	Standardizing the Use of Propofol for Washout Procedures in Critically Ill Children
504	CS	Joe Danica Inigo	Provider Attitudes on Implementing Electronic Patient-Reported Outcomes for Remote Symptom Monitoring Within Clinic Workflows: Protocol Report
505	CS	Joshua Zachariah	10-Year Trends in Transcatheter Aortic Valve Replacement (TAVR) Patient Characteristics: Insights from a Large Academic Medical Center
506	CS	Ismail Khaderi	Comparative Evaluation of Two Nasal Absorptive Sampling Devices for Patient Comfort and Cytokine Profiling in Chronic Rhinosinusitis

#	Topic	Presenter	Title
507	CS	Karthik Eswar	Development of a Computational Technique to Characterize the Architecture of Tumor-Draining Lymph Nodes in Pancreatic Ductal Adenocarcinoma
508	CS	Meera Patel	Evaluating Patient-Report Pain in Metastatic Cancer Patients
509	CS	Annette Varghese	Evaluation of an Optimal Time to Assessment after Initiation of a Food Elimination Diet in Patients with Eosinophilic Esophagitis
510	CS	Elizabeth Zhang	HIV PrEP Screening and Persistence in Breastfeeding Women with STI Symptoms in Lilongwe, Malawi
511	CS	Kelsey Lewis	Impact of Telehealth-Based Triage on Diagnostic Timeliness and Equity in Prostate Cancer Evaluation
512	CS	Sarah Bell	Integrating Fertility Markers with Cardiovascular Risk Stratification in Individuals with Turner Syndrome
513	CS	Kabir Pathak	Non-central Choroidal dynamics in Birdshot Chorioretinopathy with manual and automated analysis
514	CS	Bhamini Vellanki	Patient- and Encounter-Level Determinants and Practice-Level Variation in Emergency Department Transfer and Hospital Admission Following Urgent Care Visits in North Carolina
515	CS	Neha Das	Predictors of Rapid Progression to Severe Preeclampsia in Mild Hypertensive Disorders of Pregnancy
516	CS	Jacob Callaway	Reduced Healthcare Utilization After Perioperative Suzetrigine in Spinal Fusion and Instrumentation
517	CS	William Parker	Association Between ACEi/ARB Use and Progression of Cardiovascular-Kidney-Metabolic Syndrome in the ARIC Study
518	CS	Laughlin Ryan	Changes in Body Composition Following Weight Regain in Adults: A Systematic Review and Meta-analysis
519	CS	Sriya Dhupati	Compounded Vulnerability: Incident Sarcopenia in Patients with ESRD undergoing Bariatric Surgery
520	CS	Julie Nguyen	Evaluating the association between cardiovascular disease (CVD) risk factors and arterial stiffness among patients with chest pain in the emergency department
521	CS	Zachary Imseis	Examining Social Vulnerability as a Predictor of In-Hospital Mortality Following Blunt Thoracic Aortic Injury
522	CS	Janna Rains	Identifying Modifiable Barriers and Facilitators to Implementing Point-of-Care Ultrasound in Primary Care Offices: A CFIR-Guided Study
523	CS	Sneha A. Makhijani	The PDA Randomized Clinical Trial: Expectant Management vs Medication for Patent Ductus Arteriosus in Preterm Infants
524	CS	Emma King	Spectrum of Pathology-Confirmed Gastrointestinal Malignancies at Kamuzu Central Hospital, Malawi, 2011-2023
526	CS	Emma Roberts	Association Between Behavioral Health Service Engagement and Diabetes Control Amongst Migrant and Seasonal Agricultural Workers in Eastern North Carolina
601	CS	Jennifer Quinn	Bradycardic Collapse After Rate Control of Atrial Fibrillation in Sepsis: Beta-Blocker Toxicity, BRASH Syndrome, or Unmasked Tachy-Brady Syndrome?
602	CS	Tessa Grace Buscher	Comorbidity, Treatment Intensity, and Outcomes of Neoadjuvant Pembrolizumab-Based Chemoimmunotherapy in Older Adults With Early Triple-Negative Breast Cancer

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603	CS	Varun Indugula	Evaluating Patient Screening for an Anti-Tau Clinical Trial for Early-Onset Alzheimer's Disease
604	CS	Madison Headen	Evaluating the Prevalence of Dysphagia and Adaptive Eating Behaviors in Patients with Allergic Disease: Clues into Undiagnosed Eosinophilic Esophagitis
605	CS	Alexander Irmscher	Field-Based Assessment of Exercise-Induced Bronchoconstriction in UNC-CH Football Players with Asthma
606	CS	Mitchell Millett Kent	Impact of Concomitant Eosinophilic Asthma on Eosinophilic Esophagitis Severity and Treatment Response
607	CS	Jessica Lee	Clinical Characteristics and Treatment Outcomes of Reactive Cutaneous Squamous Cell Carcinomas and Keratoacanthomas
608	CS	Abigail Pitcairn	Primary vs Revision Anterior Cruciate Ligament Strength Reported Outcomes
609	CS	Mohanapriya Cumaran	Who Receives TAVR? Exploring Clinical and Healthcare Access Factors in Patients With Severe Aortic Stenosis
610	CS	Ananya Saravanan	Breslow Thickness and Microsatellite Detection in Primary Cutaneous Melanoma: Implications for Oncologic Surgeons and Patients
611	CS	Sai Harshith Rachakonda	Association of PSA Density and MRI Phenotype with Clinically Significant Prostate Cancer in African American men with Ambiguous PIRADS-3 Lesions
612	CS	Pranav Buneti	Age–Sex–Mechanism Transitions in Emergency Department–Treated Elbow Fractures: A National All-Ages NEISS Study (2016–2025)
613	CS	Jordan Swandell	Increasing Physical Activity Goal Setting Through Community Asset Mapping in a Rural Federally Qualified Health Center
614	CS	Emily Summers	Patient Preferences for Surgical Procedures in Hidradenitis Suppurativa: Discrete Choice Experiment
615	CS	Kaila Moore	PRDM16 Deletion Is Associated With Sex-dependent Cardiomyopathy and Cardiac Mortality: A translational, Multi-Institutional Cohort Study
616	CS	Erik Rundquist	Survival and Predictors of Mortality in Patients with an Injury Severity Score >40: A TQIP Analysis
617	CS	Meghan Stuart	CONVERT: Conversion of Nonoperative, Venous, and Extraneous Rescue Tourniquet
618	CS	Joseph Dalola	Perioperative Suzetrigine Prescription is Associated with Reduced Healthcare Utilization and Opioid Prescriptions Following Primary Total Knee Arthroplasty
619	CS	Danielle Cannon	Preprocedural Echocardiographic Predictors of Acute Right Ventricular Dysfunction Following Transcatheter Tricuspid Valve Replacement with the EVOQUE System
620	CS	Megan Little	Characterizing Maternal Interfacility Transports and Outcomes at UNC Medical Center
621	CS	Valeria Alyssa Donoso	Medication Adherence After Incarceration: Factors Associated with Missed Medication Doses Among NC FIT Participants


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